

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (ISE) (Credit System) Degree Examinations**

May - June 2023

20IS604 – MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a) Define Machine Learning. With suitable examples, explain well posed learning problems.
- b) Apply candidate elimination algorithm for the given dataset.

10 L*2 1 1

Ex.	Eyes	Nose	Head	Color	Hair	Smile
1	Round	Triangle	Round	Purple	Yes	Yes
2	Square	Square	Square	Green	Yes	No
3	Square	Triangle	Round	Yellow	Yes	Yes
4	Round	Triangle	Round	Green	No	No
5	Square	Square	Round	Yellow	Yes	Yes

10 3 1 2

2. a) With examples, illustrate the various stages of designing a learning system.
- b) With an example, illustrate K-Nearest Neighbor rule. Also explain how you handle irrelevant attributes and scaling problems.
3. a) Highlight the perspectives and issues in machine learning. With an example, explain the version space algorithm.
- b) Apply ID3 algorithm to solve the given dataset. Draw the decision tree for the same.

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10 2 2 2

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Age	Competition	Type	Profit
Old	Yes	Software	Down
Old	No	Software	Down
Old	No	Hardware	Down
Mid	Yes	Software	Down
Mid	Yes	Hardware	Down
Mid	No	Hardware	Up
Mid	No	Software	Up
New	Yes	Software	Up
New	No	Hardware	Up
New	No	Software	Up

10 3 2 2

Unit – II

4. a) Describe the challenges in unsupervised learning. With the help of an example, explain cluster analysis.
- b) With the help of an example, explain the following terms.
 - i. Error rate and classification accuracy
 - ii. Precision and recall
 - iii. Sensitivity and Specificity
5. a) Why feature maps are used ? With an example, explain the basic algorithm of self-organizing feature map.

10 2 3 1

10 2 4 1

10 2 3 1

P.T.O.

b)	With the help of examples, explain the following.				
	i. N-Fold cross validation				
	ii. AUC-ROC Curve in Machine Learning	10	2	4	1
6. a)	With suitable examples, illustrate the following.				
	i. K-Means algorithm				
	ii. Hierarchical Aggregation	12	2	3	1
b)	With a help of a diagram, illustrate learning curves and computational costs.	8	2	4	2
Unit – III					
7. a)	With the help of a diagram, explain the working principle of backpropagation algorithm.	10	2	5	2
b)	What is ensemble learning? Explain the steps to perform bagging. Also, outline the variations on the boosting theme.	10	2	5	2
8. a)	Describe the appropriate problems for artificial neural networks. With the help of a diagram, explain perceptron training rules.	10	2	5	2
b)	List and explain few instructive applications of machine learning.	10	1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
